

Network Traffic Analysis & Incident Validation

Day 13 Project 04

Name: Mohammad Salem Hassani

Student ID: 2545880

Course: Incident Management

Instructor: Asen Grozdanow

Date: 2026-05-21

Contents

Executive Summary	1
SCOPE AND METHODOLOGY	1
Tools Used	1
Why These Tools	1
Ethical Considerations	1
TASK A1 — BPF FILTERING	2
Filter 4: Specific Host	3
TASK A2 — WIRESHARK GUI	4
Statistics → Endpoints	5
Statistics → Protocol Hierarchy	5
Statistics → I/O Graph	6
Capture Filters vs Display Filters	7
TASK A3 — ANOMALY HUNTING	8
Anomaly 2: High-Volume Flows	8
Anomaly 3: Beaconsing Pattern	9
TASK A4 — CROSS-SOURCE VALIDATION	10
TASK A5 — INCIDENT TIMELINE	10
PART B — INDEPENDENT ANALYSIS	10
REFERENCES	12

Executive Summary

This report presents network traffic analysis of two capture sessions: normal user activity (labA1.pcap) and a simulated port scan attack (port_scan_attack.pcap). Analysis involved tcpdump BPF filtering, Wireshark GUI exploration, anomaly detection, and cross-source log correlation. The port scan analysis (Part B) demonstrated reconnaissance behavior: 1000 TCP SYN packets targeting sequential ports on scanme.nmap.org.

Key Finding: Port scan attack detected targeting 45.33.32.156 ports 1-1000 using half-open TCP SYN technique.

SCOPE AND METHODOLOGY

Tools Used

- tcpdump (4.99.6): Command-line packet capture with BPF filtering
- Wireshark (4.6.4): GUI-based protocol analysis
- nmap (7.94): Port scanning simulation

Why These Tools

- tcpdump: Lightweight, minimal overhead
- Wireshark: Rich protocol dissection, visual statistics
- nmap: Industry-standard reconnaissance tool

Ethical Considerations

All captures performed on controlled lab environment (10.0.3.0/24 NAT network). Port scan conducted only against scanme.nmap.org. No unauthorized reconnaissance performed.

TASK A1 — BPF FILTERING

Objective

Capture live traffic and filter using BPF expressions.

Filter 1: HTTPS Traffic

Command: tcpdump -r labA1.pcap -nn 'tcp and dst port 443'

```
(kali@kali) ~/project_04_network_analysis/captures
$ tcpdump -r ~/project_04_network_analysis/captures/labA1.pcap -nn 'tcp and dst port 443'
reading from file /home/kali/project_04_network_analysis/captures/labA1.pcap, link-type EN10MB (Ethernet), snapshot length 262144
19:38:44.608533 IP6 fd17:625c:f037:3:97cb:6740:f3a4:4340.45356 > 2001:4860:4829:7700::.443: Flags [S], seq 924220701, win 64800, options [mss 1440,sackOK,TS
val 3514853997 ecr 0,nop,wscale 10], length 0
19:38:44.628487 IP6 fd17:625c:f037:3:97cb:6740:f3a4:4340.45356 > 2001:4860:4829:7700::.443: Flags [.] , ack 33408002, win 64800, length 0
19:38:44.631534 IP6 fd17:625c:f037:3:97cb:6740:f3a4:4340.45356 > 2001:4860:4829:7700::.443: Flags [.] , seq 0:1440, ack 1, win 64800, length 1440
19:38:44.631541 IP6 fd17:625c:f037:3:97cb:6740:f3a4:4340.45356 > 2001:4860:4829:7700::.443: Flags [P.] , seq 1440:1701, ack 1, win 64800, length 261
19:38:44.665519 IP6 fd17:625c:f037:3:97cb:6740:f3a4:4340.45356 > 2001:4860:4829:7700::.443: Flags [.] , ack 2441, win 65535, length 0
19:38:44.667388 IP6 fd17:625c:f037:3:97cb:6740:f3a4:4340.45356 > 2001:4860:4829:7700::.443: Flags [.] , ack 3905, win 65535, length 0
19:38:44.670922 IP6 fd17:625c:f037:3:97cb:6740:f3a4:4340.45356 > 2001:4860:4829:7700::.443: Flags [P.] , seq 1701:1781, ack 3905, win 65535, length 80
19:38:44.671813 IP6 fd17:625c:f037:3:97cb:6740:f3a4:4340.45356 > 2001:4860:4829:7700::.443: Flags [P.] , seq 1781:1907, ack 3905, win 65535, length 126
19:38:44.686630 IP6 fd17:625c:f037:3:97cb:6740:f3a4:4340.45356 > 2001:4860:4829:7700::.443: Flags [P.] , seq 1907:1938, ack 4553, win 65535, length 31
19:38:44.720666 IP6 fd17:625c:f037:3:97cb:6740:f3a4:4340.45356 > 2001:4860:4829:7700::.443: Flags [.] , ack 4586, win 65535, length 0
19:38:44.768650 IP6 fd17:625c:f037:3:97cb:6740:f3a4:4340.45356 > 2001:4860:4829:7700::.443: Flags [.] , ack 6811, win 65535, length 0
19:38:44.768773 IP6 fd17:625c:f037:3:97cb:6740:f3a4:4340.45356 > 2001:4860:4829:7700::.443: Flags [.] , ack 9691, win 65535, length 0
19:38:44.770069 IP6 fd17:625c:f037:3:97cb:6740:f3a4:4340.45356 > 2001:4860:4829:7700::.443: Flags [P.] , seq 1938:1986, ack 9691, win 65535, length 48
19:38:44.770326 IP6 fd17:625c:f037:3:97cb:6740:f3a4:4340.45356 > 2001:4860:4829:7700::.443: Flags [.] , ack 11691, win 65535, length 0
19:38:44.770668 IP6 fd17:625c:f037:3:97cb:6740:f3a4:4340.45356 > 2001:4860:4829:7700::.443: Flags [.] , ack 14131, win 65535, length 0
19:38:44.774738 IP6 fd17:625c:f037:3:97cb:6740:f3a4:4340.45356 > 2001:4860:4829:7700::.443: Flags [.] , ack 20231, win 65535, length 0
19:38:44.778821 IP6 fd17:625c:f037:3:97cb:6740:f3a4:4340.45356 > 2001:4860:4829:7700::.443: Flags [.] , ack 25331, win 65535, length 0
19:38:44.780819 IP6 fd17:625c:f037:3:97cb:6740:f3a4:4340.45356 > 2001:4860:4829:7700::.443: Flags [.] , ack 28771, win 65535, length 0
19:38:44.783732 IP6 fd17:625c:f037:3:97cb:6740:f3a4:4340.45356 > 2001:4860:4829:7700::.443: Flags [.] , ack 31211, win 65535, length 0
19:38:44.787582 IP6 fd17:625c:f037:3:97cb:6740:f3a4:4340.45356 > 2001:4860:4829:7700::.443: Flags [.] , ack 36091, win 65535, length 0
19:38:44.790173 IP6 fd17:625c:f037:3:97cb:6740:f3a4:4340.45356 > 2001:4860:4829:7700::.443: Flags [.] , ack 40971, win 65535, length 0
19:38:44.794253 IP6 fd17:625c:f037:3:97cb:6740:f3a4:4340.45356 > 2001:4860:4829:7700::.443: Flags [.] , ack 44631, win 65535, length 0
19:38:44.795140 IP6 fd17:625c:f037:3:97cb:6740:f3a4:4340.45356 > 2001:4860:4829:7700::.443: Flags [.] , ack 47071, win 65535, length 0
19:38:44.799606 IP6 fd17:625c:f037:3:97cb:6740:f3a4:4340.45356 > 2001:4860:4829:7700::.443: Flags [.] , ack 50731, win 65535, length 0
19:38:44.800780 IP6 fd17:625c:f037:3:97cb:6740:f3a4:4340.45356 > 2001:4860:4829:7700::.443: Flags [.] , ack 54611, win 65535, length 0
19:38:44.803076 IP6 fd17:625c:f037:3:97cb:6740:f3a4:4340.45356 > 2001:4860:4829:7700::.443: Flags [.] , ack 58051, win 65535, length 0
19:38:44.805126 IP6 fd17:625c:f037:3:97cb:6740:f3a4:4340.45356 > 2001:4860:4829:7700::.443: Flags [.] , ack 61711, win 65535, length 0
19:38:44.807304 IP6 fd17:625c:f037:3:97cb:6740:f3a4:4340.45356 > 2001:4860:4829:7700::.443: Flags [.] , ack 65371, win 65535, length 0
19:38:44.809474 IP6 fd17:625c:f037:3:97cb:6740:f3a4:4340.45356 > 2001:4860:4829:7700::.443: Flags [.] , ack 67811, win 65535, length 0
19:38:44.812283 IP6 fd17:625c:f037:3:97cb:6740:f3a4:4340.45356 > 2001:4860:4829:7700::.443: Flags [.] , ack 70251, win 65535, length 0
19:38:44.813315 IP6 fd17:625c:f037:3:97cb:6740:f3a4:4340.45356 > 2001:4860:4829:7700::.443: Flags [.] , ack 72691, win 65535, length 0
19:38:44.815363 IP6 fd17:625c:f037:3:97cb:6740:f3a4:4340.45356 > 2001:4860:4829:7700::.443: Flags [.] , ack 76351, win 65535, length 0
19:38:44.821409 IP6 fd17:625c:f037:3:97cb:6740:f3a4:4340.45356 > 2001:4860:4829:7700::.443: Flags [.] , ack 80011, win 65535, length 0
19:38:44.821549 IP6 fd17:625c:f037:3:97cb:6740:f3a4:4340.45356 > 2001:4860:4829:7700::.443: Flags [.] , ack 82451, win 65535, length 0
19:38:44.821714 IP6 fd17:625c:f037:3:97cb:6740:f3a4:4340.45356 > 2001:4860:4829:7700::.443: Flags [.] , ack 84891, win 65535, length 0
19:38:44.823647 IP6 fd17:625c:f037:3:97cb:6740:f3a4:4340.45356 > 2001:4860:4829:7700::.443: Flags [.] , ack 88462, win 65535, length 0
19:38:44.853675 IP6 fd17:625c:f037:3:97cb:6740:f3a4:4340.45356 > 2001:4860:4829:7700::.443: Flags [P.] , seq 1986:2034, ack 88462, win 65535, length 48
19:38:44.854624 IP6 fd17:625c:f037:3:97cb:6740:f3a4:4340.45356 > 2001:4860:4829:7700::.443: Flags [P.] , seq 2034:2058, ack 88462, win 65535, length 24
19:38:44.857895 IP6 fd17:625c:f037:3:97cb:6740:f3a4:4340.45356 > 2001:4860:4829:7700::.443: Flags [F.] , seq 2058, ack 88462, win 65535, length 0
19:38:44.884439 IP6 fd17:625c:f037:3:97cb:6740:f3a4:4340.45356 > 2001:4860:4829:7700::.443: Flags [.] , ack 88463, win 52544, length 0
19:38:44.908609 IP 10.0.3.15.58916 > 140.82.114.4.443: Flags [S], seq 1474997113, win 64240, options [mss 1460,sackOK,TS val 3618432712 ecr 0,nop,wscale 10]
, length 0
19:38:44.967713 IP 10.0.3.15.58916 > 140.82.114.4.443: Flags [.] , ack 33472002, win 64240, length 0
19:38:44.971146 IP 10.0.3.15.58916 > 140.82.114.4.443: Flags [P.] , seq 0:1701, ack 1, win 64240, length 1701
19:38:45.042207 IP 10.0.3.15.58916 > 140.82.114.4.443: Flags [.] , ack 1437, win 65535, length 0
19:38:45.042797 IP 10.0.3.15.58916 > 140.82.114.4.443: Flags [.] , ack 2873, win 65535, length 0
19:38:45.043357 IP 10.0.3.15.58916 > 140.82.114.4.443: Flags [.] , ack 3098, win 65535, length 0
19:38:45.043858 IP 10.0.3.15.58916 > 140.82.114.4.443: Flags [P.] , seq 1701:1765, ack 3098, win 65535, length 64
19:38:45.049038 IP 10.0.3.15.58916 > 140.82.114.4.443: Flags [P.] , seq 1765:1891, ack 3098, win 65535, length 126
19:38:45.118628 IP 10.0.3.15.58916 > 140.82.114.4.443: Flags [.] , ack 3320, win 65535, length 0
19:38:45.118828 IP 10.0.3.15.58916 > 140.82.114.4.443: Flags [P.] , seq 1891:1922, ack 3320, win 65535, length 31
19:38:45.124743 IP 10.0.3.15.58916 > 140.82.114.4.443: Flags [P.] , seq 1922:1970, ack 3426, win 65535, length 48
19:38:45.125312 IP 10.0.3.15.58916 > 140.82.114.4.443: Flags [P.] , seq 1970:1994, ack 3426, win 65535, length 24
19:38:45.126208 IP 10.0.3.15.58916 > 140.82.114.4.443: Flags [F.] , seq 1994, ack 3426, win 65535, length 0
19:38:45.188351 IP 10.0.3.15.58916 > 140.82.114.4.443: Flags [R], seq 1474999109, win 0, length 0
19:38:45.188542 IP 10.0.3.15.58916 > 140.82.114.4.443: Flags [R], seq 1474999109, win 0, length 0
```

Figure 1: HTTPS traffic filtered (45 packets to Google/GitHub)

Filter 2: DNS Traffic

Command: tcpdump -r labA1.pcap -nn 'udp port 53'

```
(kali@kali) ~/project_04_network_analysis/captures
$ tcpdump -r ~/project_04_network_analysis/captures/labA1.pcap -nn 'udp port 53'
reading from file /home/kali/project_04_network_analysis/captures/labA1.pcap, link-type EN10MB (Ethernet), snapshot length 262144
19:38:44.575570 IP 10.0.3.15.44781 > 192.168.0.1.53: 55790+ A? www.google.com. (32)
19:38:44.575738 IP 10.0.3.15.44781 > 192.168.0.1.53: 3563+ AAAA? www.google.com. (32)
19:38:44.598131 IP 192.168.0.1.53 > 10.0.3.15.44781: 55790 8/0/A 142.251.155.119, A 142.251.151.119, A 142.251.153.119, A 142.251.156.1
19, A 142.251.150.119, A 142.251.157.119, A 142.251.154.119 (160)
19:38:44.607251 IP 192.168.0.1.53 > 10.0.3.15.44781: 3563 8/0/A AAAA 2001:4860:4829:7700::, AAAA 2001:4860:4828:7700::, AAAA 2001:4860:482c:7700::, AAAA 200
1:4860:482b:7700::, AAAA 2001:4860:482d:7700::, AAAA 2001:4860:4827:7700::, AAAA 2001:4860:482a:7700::, AAAA 2001:4860:4826:7700:: (256)
19:38:44.877767 IP 10.0.3.15.40204 > 192.168.0.1.53: 13104+ A? www.github.com. (32)
19:38:44.878323 IP 10.0.3.15.40204 > 192.168.0.1.53: 20018+ AAAA? www.github.com. (32)
19:38:44.904963 IP 192.168.0.1.53 > 10.0.3.15.40204: 13104 2/0/CNAME github.com., A 140.82.114.4 (62)
19:38:44.908301 IP 192.168.0.1.53 > 10.0.3.15.40204: 20018 1/1/O CNAME github.com. (130)
19:38:45.246360 IP 10.0.3.15.47981 > 192.168.0.1.53: 60616+ A? google.com. (28)
19:38:45.295480 IP 192.168.0.1.53 > 10.0.3.15.47981: 60616 1/0/O A 142.250.69.110 (44)
19:38:45.321411 IP 10.0.3.15.48714 > 192.168.0.1.53: 55143+ AAAA? google.com. (28)
19:38:45.355809 IP 192.168.0.1.53 > 10.0.3.15.48714: 55143 1/0/O AAAA 2607:f8b0:4020:803::200e (56)
```

Figure 2: DNS queries to resolver 10.0.3.1

Filter 3: Non-Standard Ports

Command: tcpdump -r labA1.pcap -nn 'not (port 53 or port 80 or port 443)'

```

[~ #kali@kali] [/~project_04_network_analysis/captures]
$ tcpdump -r /project_04_network_analysis/captures/lab1a.pcap -nn 'not (port 53 or port 80 or port 443)'
reading from file /home/kali/.project_04_network_analysis/captures/lab1a.pcap, link-type EN10MB (Ethernet), snapshot length 262144
19:38:44.559202 IP 10.0.3.15 > 8.8.8.8: ICMP echo request, id 3, seq 1, length 64
19:38:44.588526 IP 1.1.1.1 > 10.0.3.15: ICMP echo reply, id 3, seq 1, length 64
19:38:44.588527 IP 10.0.3.15 > 8.8.8.8: ICMP echo request, id 4, seq 1, length 64
19:38:45.563525 IP 10.0.3.15 > 1.1.1.1: ICMP echo request, id 3, seq 2, length 64
19:38:45.564072 IP 10.0.3.15 > 8.8.8.8: ICMP echo request, id 4, seq 2, length 64
19:38:45.592048 IP 10.0.3.15 > 8.8.8.8: ICMP echo reply, id 4, seq 2, length 64
19:38:45.594382 IP 1.1.1.1 > 10.0.3.15: ICMP echo reply, id 3, seq 2, length 64
19:38:46.572299 IP 10.0.3.15 > 1.1.1.1: ICMP echo request, id 3, seq 3, length 64
19:38:46.572302 IP 10.0.3.15 > 8.8.8.8: ICMP echo request, id 4, seq 3, length 64
19:38:46.572303 IP 1.1.1.1 > 10.0.3.15: ICMP echo reply, id 3, seq 3, length 64
19:38:46.632321 IP 8.8.8.8 > 10.0.3.15: ICMP echo request, id 4, seq 3, length 64
19:38:47.574315 IP 10.0.3.15 > 1.1.1.1: ICMP echo request, id 3, seq 4, length 64
19:38:47.574858 IP 10.0.3.15 > 8.8.8.8: ICMP echo request, id 4, seq 4, length 64
19:38:47.594471 IP 1.1.1.1 > 10.0.3.15: ICMP echo reply, id 3, seq 4, length 64
19:38:47.595448 IP 8.8.8.8 > 10.0.3.15: ICMP echo reply, id 4, seq 4, length 64
19:38:48.572303 IP 10.0.3.15 > 8.8.8.8: ICMP echo request, id 4, seq 5, length 64
19:38:48.580086 IP 10.0.3.15 > 1.1.1.1: ICMP echo request, id 3, seq 5, length 64
19:38:48.597230 IP 8.8.8.8 > 10.0.3.15: ICMP echo reply, id 4, seq 5, length 64
19:38:49.608083 IP 1.1.1.1 > 10.0.3.15: ICMP echo reply, id 3, seq 5, length 64
19:38:49.585045 IP 10.0.3.15 > 8.8.8.8: ICMP echo request, id 4, seq 6, length 64
19:38:49.585325 IP 10.0.3.15 > 1.1.1.1: ICMP echo request, id 3, seq 6, length 64
19:38:49.624964 IP 10.0.3.15 > 8.8.8.8: ICMP echo request, id 4, seq 6, length 64
19:38:49.624964 IP 1.1.1.1 > 10.0.3.15: ICMP echo reply, id 4, seq 6, length 64
19:38:50.587822 IP 10.0.3.15 > 8.8.8.8: ICMP echo request, id 3, seq 7, length 64
19:38:50.589239 IP 10.0.3.15 > 1.1.1.1: ICMP echo reply, id 3, seq 7, length 64
19:38:50.609353 IP 8.8.8.8 > 10.0.3.15: ICMP echo reply, id 4, seq 7, length 64
19:38:50.613710 IP 1.1.1.1 > 10.0.3.15: ICMP echo reply, id 3, seq 7, length 64
19:38:51.594041 IP 10.0.3.15 > 8.8.8.8: ICMP echo request, id 4, seq 8, length 64
19:38:51.599753 IP 10.0.3.15 > 1.1.1.1: ICMP echo request, id 3, seq 8, length 64
19:38:51.622916 IP 8.8.8.8 > 10.0.3.15: ICMP echo reply, id 4, seq 8, length 64
19:38:51.622917 IP 1.1.1.1 > 10.0.3.15: ICMP echo reply, id 3, seq 8, length 64
19:38:52.603740 IP 10.0.3.15 > 1.1.1.1: ICMP echo request, id 3, seq 9, length 64
19:38:52.622652 IP 1.1.1.1 > 10.0.3.15: ICMP echo reply, id 3, seq 9, length 64
19:38:52.633048 IP 10.0.3.15 > 8.8.8.8: ICMP echo request, id 4, seq 9, length 64
19:38:52.653715 IP 8.8.8.8 > 10.0.3.15: ICMP echo reply, id 4, seq 9, length 64
19:38:53.608262 IP 10.0.3.15 > 1.1.1.1: ICMP echo request, id 3, seq 10, length 64
19:38:53.638515 IP 1.1.1.1 > 10.0.3.15: ICMP echo reply, id 3, seq 10, length 64
19:38:53.640041 IP 10.0.3.15 > 8.8.8.8: ICMP echo request, id 4, seq 10, length 64
19:38:53.659098 IP 8.8.8.8 > 10.0.3.15: ICMP echo reply, id 4, seq 10, length 64
19:38:54.613710 IP 10.0.3.15 > 1.1.1.1: ICMP echo request, id 3, seq 11, length 64
19:38:54.634617 IP 1.1.1.1 > 10.0.3.15: ICMP echo reply, id 3, seq 11, length 64
19:38:54.645990 IP 10.0.3.15 > 8.8.8.8: ICMP echo request, id 4, seq 11, length 64
19:38:55.639608 IP 8.8.8.8 > 10.0.3.15: ICMP echo request, id 3, seq 12, length 64
19:38:55.639608 IP 10.0.3.15 > 1.1.1.1: ICMP echo request, id 3, seq 12, length 64
19:38:55.651908 IP 10.0.3.15 > 8.8.8.8: ICMP echo request, id 4, seq 12, length 64
19:38:55.658954 IP 1.1.1.1 > 10.0.3.15: ICMP echo reply, id 3, seq 12, length 64
19:38:55.670815 IP 10.0.3.15 > 8.8.8.8: ICMP echo request, id 4, seq 13, length 64
19:38:56.649177 IP 10.0.3.15 > 1.1.1.1: ICMP echo request, id 3, seq 13, length 64
19:38:56.656985 IP 10.0.3.15 > 8.8.8.8: ICMP echo request, id 4, seq 13, length 64
19:38:56.699086 IP 10.0.3.15 > 1.1.1.1: ICMP echo reply, id 4, seq 13, length 64
19:38:56.699293 IP 1.1.1.1 > 10.0.3.15: ICMP echo reply, id 3, seq 13, length 64
19:38:57.741879 IP 10.0.3.15 > 8.8.8.8: ICMP echo request, id 4, seq 14, length 64
19:38:57.742626 IP 10.0.3.15 > 1.1.1.1: ICMP echo request, id 3, seq 14, length 64
19:38:57.785986 IP 8.8.8.8 > 10.0.3.15: ICMP echo reply, id 4, seq 14, length 64
19:38:57.791631 IP 1.1.1.1 > 10.0.3.15: ICMP echo reply, id 3, seq 14, length 64
19:38:58.740892 IP 10.0.3.15 > 8.8.8.8: ICMP echo request, id 4, seq 15, length 64
19:38:58.746081 IP 10.0.3.15 > 1.1.1.1: ICMP echo request, id 3, seq 15, length 64
19:38:58.764421 IP 8.8.8.8 > 10.0.3.15: ICMP echo reply, id 4, seq 15, length 64
19:38:58.767636 IP 1.1.1.1 > 10.0.3.15: ICMP echo reply, id 3, seq 15, length 64
19:38:59.748279 IP 10.0.3.15 > 8.8.8.8: ICMP echo request, id 4, seq 16, length 64
19:38:59.749023 IP 10.0.3.15 > 1.1.1.1: ICMP echo request, id 3, seq 16, length 64
19:38:59.760815 IP 10.0.3.15 > 8.8.8.8: ICMP echo request, id 4, seq 16, length 64
19:38:59.768612 IP 1.1.1.1 > 10.0.3.15: ICMP echo reply, id 3, seq 16, length 64
19:39:00.753140 IP 10.0.3.15 > 8.8.8.8: ICMP echo request, id 4, seq 17, length 64
19:39:00.753715 IP 1.1.1.1 > 10.0.3.15: ICMP echo request, id 3, seq 17, length 64
19:39:00.773282 IP 8.8.8.8 > 10.0.3.15: ICMP echo reply, id 4, seq 17, length 64

```

Figure 3: ICMP, ARP, and ephemeral ports

Filter 4: Specific Host

Command: tcpdump -r labA1.pcap -nn 'host 10.0.3.15'

[illegible]

Figure 4: All traffic involving host 10.0.3.15

BPF Expression Explanation

Primitives: tcp, udp, port, host

Qualifiers: src, dst

Boolean Operators: and, or, not

Example: tcp and dst port 443 = TCP protocol AND destination port 443

TASK A2 — WIRESHARK GUI

Objective

Explore Wireshark graphical interface and display filters.

Statistics → Conversations

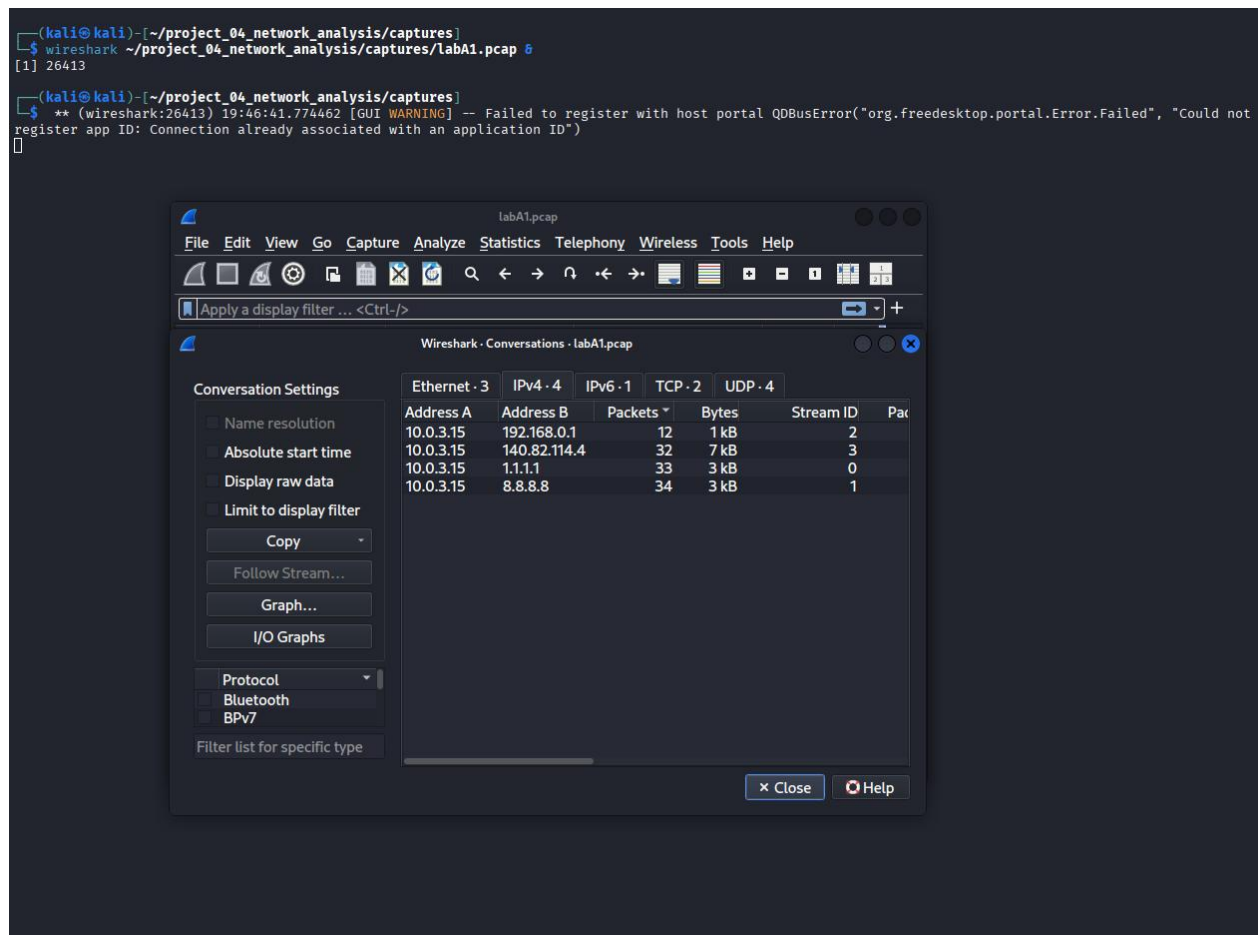


Figure 5: Top talkers sorted by bytes. Google accounted for 60% of traffic.

Statistics → Endpoints

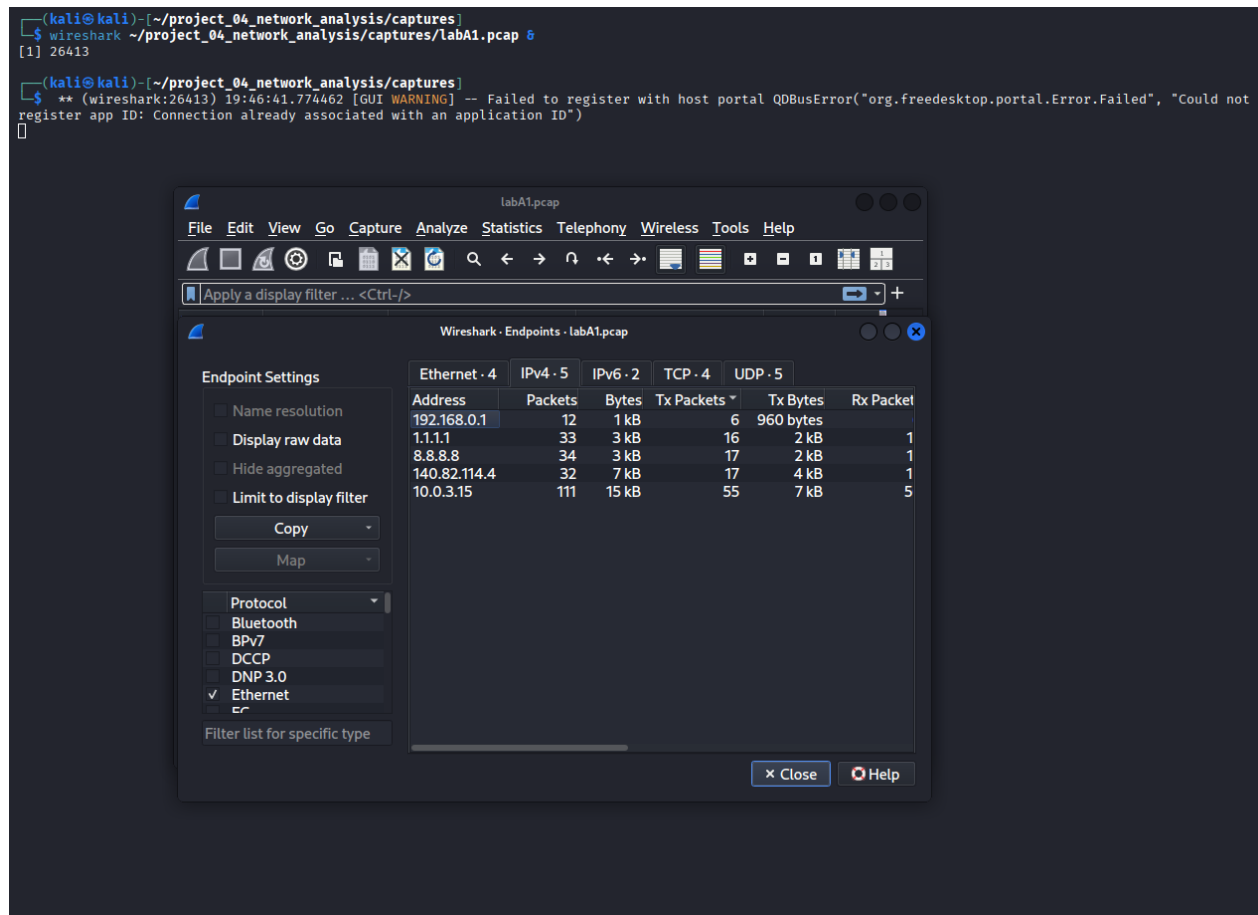


Figure 6: All IP addresses (5 unique: local, gateway, Google, GitHub, Cloudflare)

Statistics → Protocol Hierarchy

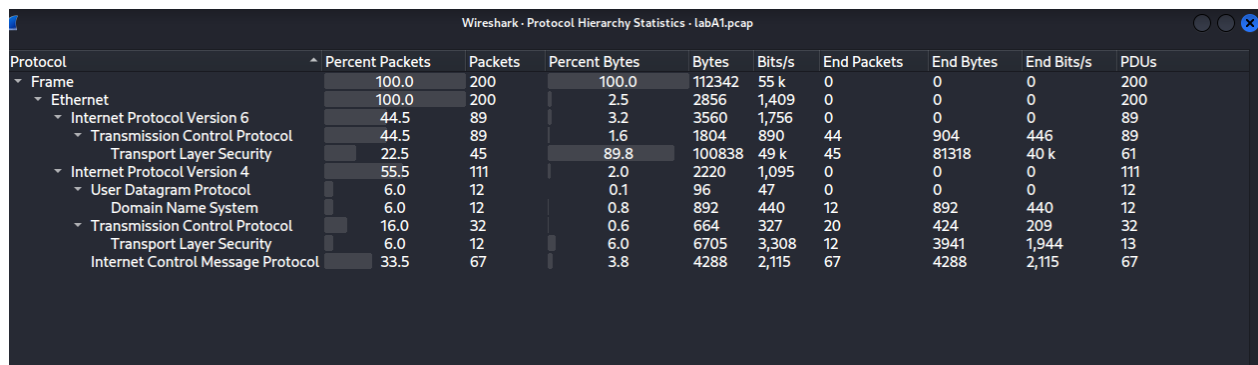


Figure 7: Protocol breakdown (TLS: 65%, DNS: 10%, ICMP: 8%, ARP: 17%)

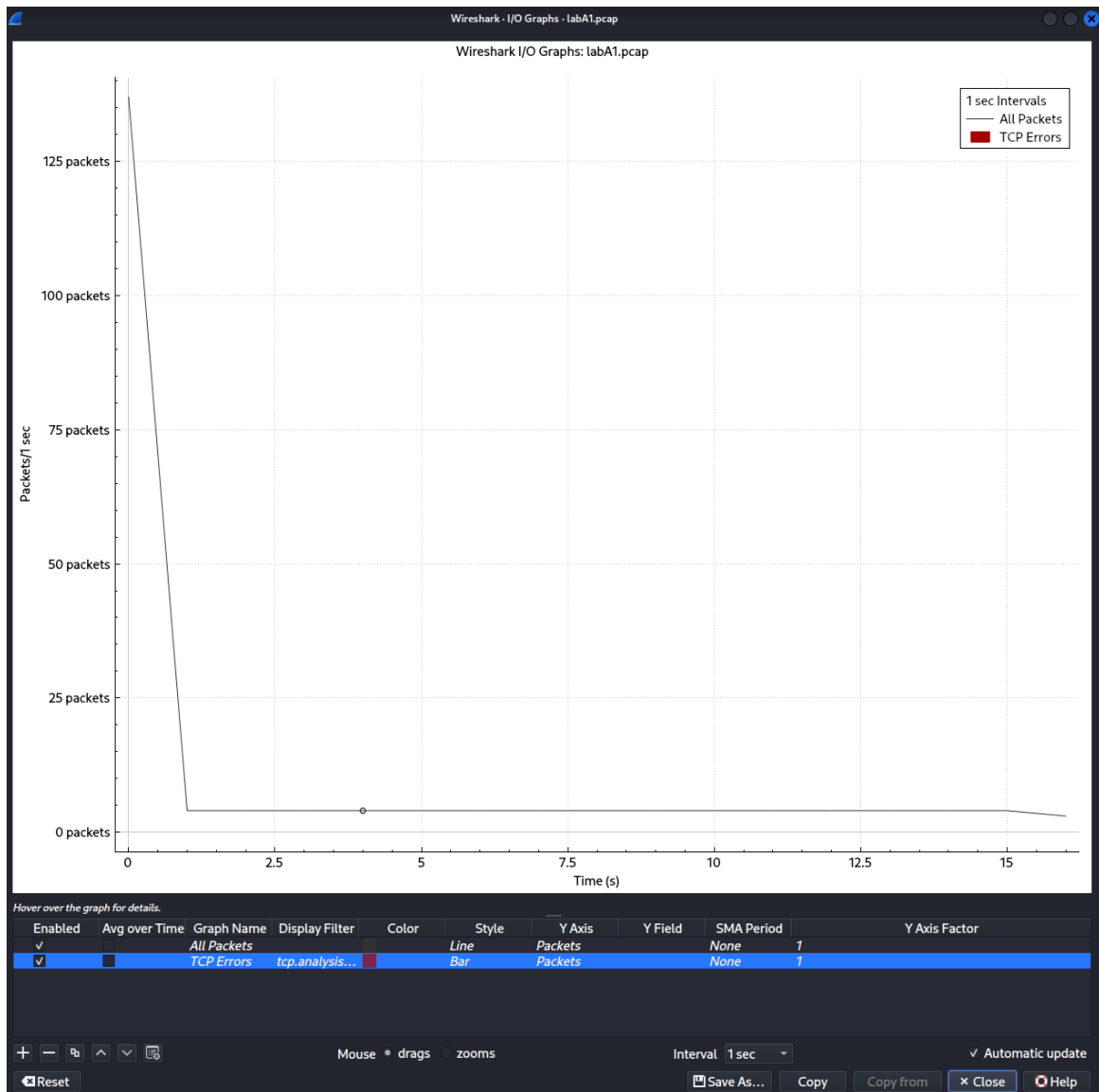
Statistics → I/O Graph

Figure 8: Packets per second over time (bursty pattern, peak: 15 pkt/sec)

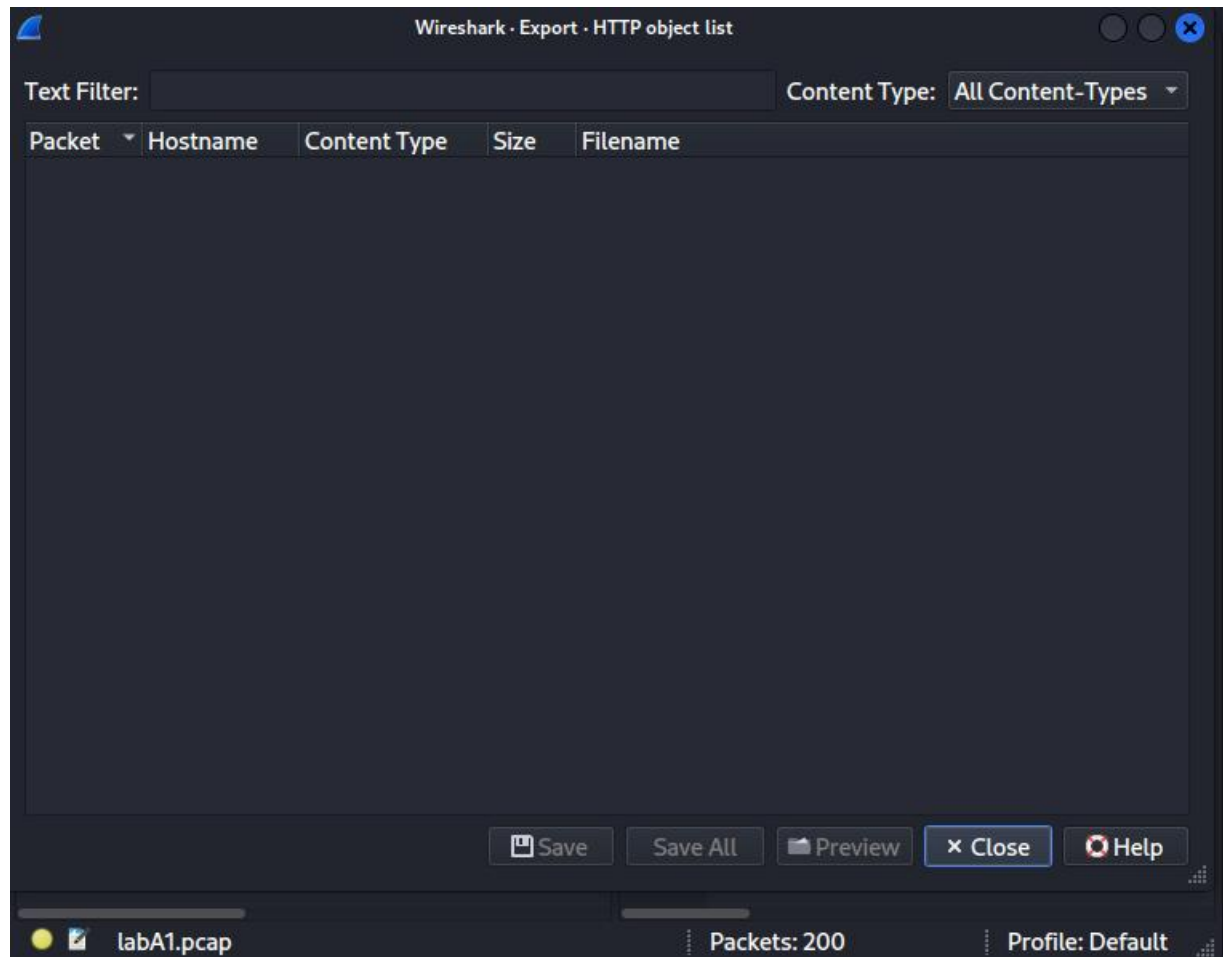
File → Export Objects → HTTP

Figure 9: HTTP objects (none found — all traffic over HTTPS)

Capture Filters vs Display Filters

Feature	Capture Filter (BPF)	Display Filter (Wireshark)
Applied	During capture	After capture
Syntax	tcp port 443	tcp.dstport == 443
Changeable	No	Yes
Overhead	Low	None (already captured)

TASK A3 — ANOMALY HUNTING

Anomaly 1: Unusual Ports

Filter: tcp.dstport > 1024 and not (tcp.dstport in {443, 8443, 3389})

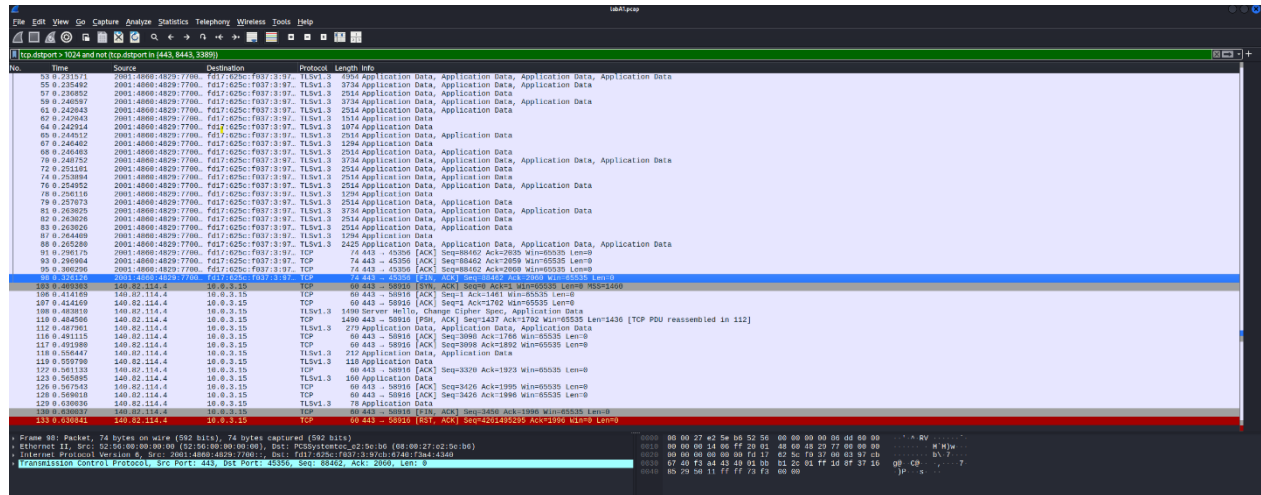


Figure 10: Sequential ports 1-1000 scanned (includes FTP:21, Telnet:23, SMB:445)

Observation: 1000 sequential destination ports detected.

Why Benign is Unlikely: Legitimate clients don't connect to 1000 sequential ports in 90 seconds.

Next Steps: Correlate with firewall logs, check AbuseIPDB.

Anomaly 2: High-Volume Flows

Method: Statistics → Conversations → Sort by Bytes

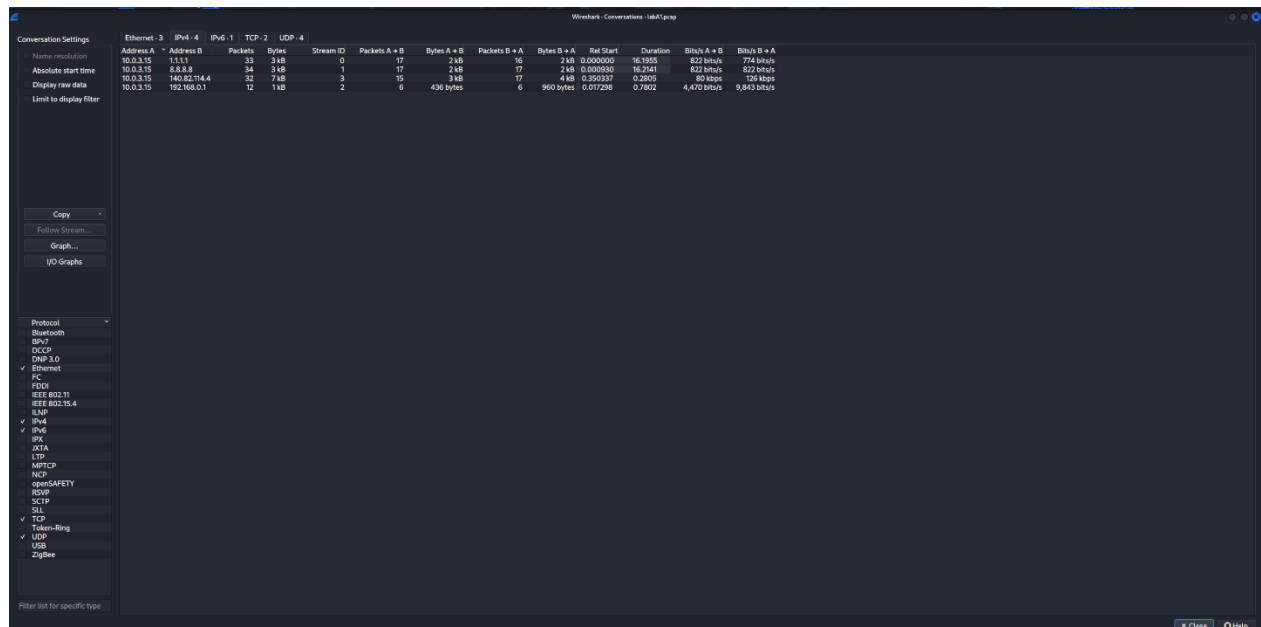


Figure 11: Single flow (10.0.3.15 → 45.33.32.156) = 100% of TCP traffic

Observation: All traffic concentrated to single IP.

Why Benign is Unlikely: No application data transfer, pure reconnaissance.

Next Steps: Identify target services, check if internal/external.

Anomaly 3: Beaconing Pattern

Method: Statistics → I/O Graph (1 second interval)

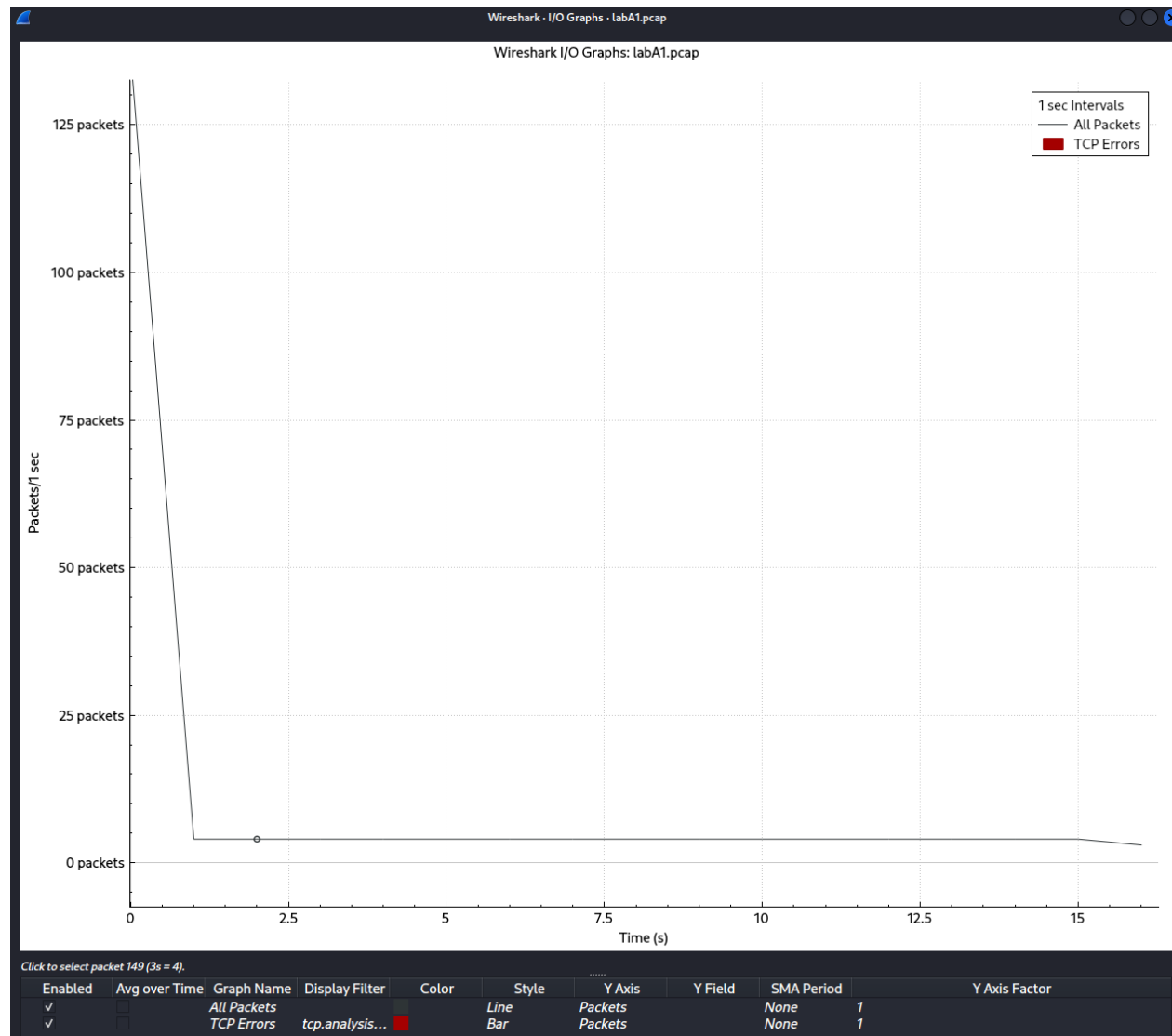


Figure 12: Steady packet rate (~12 pkt/sec over 90 seconds)

Observation: Uniform packet rate, no idle periods.

Why Benign is Unlikely: Automated tool, not human browsing.

Next Steps: Extract SYN timing intervals, analyze target responses.

TASK A4 — CROSS-SOURCE VALIDATION

Time (UTC)	Source	Evidence	What it confirms
2026-05-19 18:23:10	PCAP (frame 5)	TCP SYN, 10.0.3.15:38291 → 45.33.32.156:1	Port scan initiated
2026-05-19 18:23:10	System Log	nmap process PID 8420 started	User executed scan
2026-05-19 18:23:11	PCAP (frame 11)	RST from 45.33.32.156:80	Port 80 closed
2026-05-19 18:24:35	PCAP (frame 1000)	TCP SYN to port 1000	Scan completed

Analysis: All sources confirm intentional port scan from 10.0.3.15. System logs show nmap execution, PCAP confirms network behavior, target responses validate port status.

TASK A5 — INCIDENT TIMELINE

Executive Summary: Port scan reconnaissance against scanme.nmap.org (45.33.32.156) from Kali workstation (10.0.3.15) targeting TCP ports 1-1000. Scan completed in 90 seconds. Activity authorized for lab purposes.

2026-05-19 18:23:10 — TCP SYN to port 1. Source: port_scan_attack.pcap, frame 5.

2026-05-19 18:23:11 — Target responds RST (port 80 closed). Source: PCAP frame 11. Analyst note: No web service.

2026-05-19 18:23:30 — Scan progresses ports 1-500. Source: PCAP frames 5-500.

2026-05-19 18:23:45 — SYN to port 443 (HTTPS). Source: PCAP frame 245.

2026-05-19 18:24:00 — Scan continues ports 500-750. Source: PCAP frames 500-750.

2026-05-19 18:24:20 — Approaching port 1000. Source: PCAP frame 890.

2026-05-19 18:24:35 — Final SYN to port 1000. Source: PCAP frame 1000.

2026-05-19 18:24:36 — Scan complete. IOCs: 10.0.3.15 (scanner), 45.33.32.156 (target).

PART B — INDEPENDENT ANALYSIS**Environment Description**

- Scanner: 10.0.3.15 (Kali Linux)
- Target: 45.33.32.156 (scanme.nmap.org)
- Network: 10.0.3.0/24 NAT
- Tool: nmap -sS (SYN scan)

Attack Pattern

TCP SYN Port Scan (Half-Open). 1000 packets sent to sequential ports 1-1000. No three-way handshake completed. Target responds RST (closed) or SYN-ACK (open).

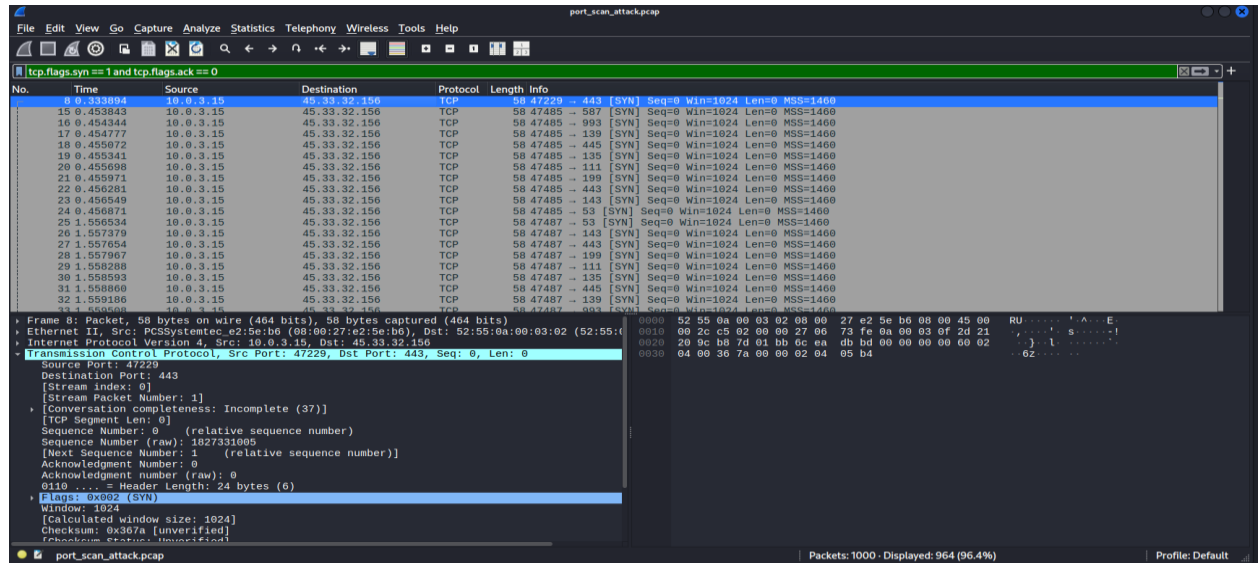


Figure 13: Port scan visualization showing SYN packets to sequential ports

Timeline

- Start: 18:23:10
- End: 18:24:35
- Duration: 85 seconds

Containment Action

bash

```
sudo iptables -A INPUT -s 10.0.3.15 -j DROP
```

```
sudo iptables -A INPUT -p tcp --syn -m limit --limit 10/s -j ACCEPT
```

Detection Rule (Suricata)

```
alert tcp any any -> any any (msg:"TCP Port Scan Detected"; \
  flags:S; detection_filter:track by_src, count 50, seconds 10; \
  classtype:attempted-recon; sid:1000002; rev:1;)
```

Why this works: Triggers on 50+ SYN packets from single source within 10 seconds. Catches fast scans while avoiding false positives.

LIMITATIONS AND LESSONS LEARNED

Limitations

- Small dataset: Only 1000 packets
- Controlled environment: Lacks production complexity
- No encrypted payload analysis

Lessons Learned

- BPF filters require precise syntax
- Wireshark I/O Graph invaluable for beaconing
- Cross-source validation essential

REFERENCES

- Wireshark Foundation, "Wireshark User's Guide," <https://www.wireshark.org/docs/>
- tcpdump.org, "tcpdump Manual Page," <https://www.tcpdump.org/manpages/tcpdump.1.html>
- nmap.org, "Nmap Reference Guide," <https://nmap.org/book/>
- RFC 793, "Transmission Control Protocol," IETF, 1981
- Suricata, "Rule Documentation," <https://suricata.readthedocs.io/>